

JDBC Theory Questions:

1. What is JDBC? What are the steps to connect Java with database?
2. What is JDBC driver? List its types.
3. What is Difference between DriverManager and Connection interface?
4. Difference between Statement and PreparedStatement?
5. What is ResultSet? What are types of ResultSet?
6. What is difference between executeQuery() and executeUpdate()?
7. What is batch processing?
8. What is transaction management?
9. What is auto-commit mode, commit() and rollback()?
10. What is PreparedStatement precompilation and What is SQL Injection?

Scenario-Based Questions

- 1 How to insert data safely?
- 2 How to handle multiple queries?
- 3 How to fetch data from DB?
- 4 How to update records?
- 5 How to delete records?

MCQ Questions

- 1 JDBC stands for:
A. Java Database Connectivity
B. Java Data Control
C. Java DB Connection
D. None
- 2 Which package is used?
A. java.sql
B. java.io
C. java.util
D. java.net
- 3 Which interface executes SQL?
A. Connection
B. Statement
C. Driver
D. ResultSet

5. Which is faster?
 - A. Statement
 - B. PreparedStatement
 - C. Both same
 - D. None

6. Which method executes SELECT?
 - A. executeUpdate()
 - B. executeQuery()
 - C. execute()
 - D. run()

- 7 Which method executes INSERT?
 - A. executeQuery()
 - B. executeUpdate()
 - C. executeSelect()
 - D. executeData()

8. ResultSet stores:
 - A. Connection
 - B. Query result
 - C. Driver
 - D. SQL

9. Default auto-commit mode?
 - A. true
 - B. false
 - C. null
 - D. depends

9. Which prevents SQL Injection?
 - A. Statement
 - B. PreparedStatement
 - C. ResultSet
 - D. Driver

10. Exception in JDBC:
 - A. IOException
 - B. SQLException
 - C. NullPointerException
 - D. ArithmeticException

Practice Programs

- 1) Write a Java program to establish a connection with a database (e.g., MySQL).

Load JDBC driver

Create connection using URL, username, password

Display message: "Connection Successful" or "Failed".

- 2) Insert a new student record into a database table.

Example Table:

```
student(id, name, marks)
```

Input:

id = 1, name = "John", marks = 85

Output:

Record inserted successfully

- 3) Insert view , update and delete records using `PreparedStatement` to avoid SQL injection.
- 4) an existing record in the database.Delete a record from the database using ID.
- 5) Insert view , update and delete records using `PreparedStatement` to avoid SQL injection.

- 6) Simulate a bank transaction between two accounts.

Table:

```
accounts(id, balance)
```

Task:

- Deduct amount from sender
- Add amount to receiver
- If any step fails → rollback
- Else → commit

- 7) Insert multiple records efficiently using batch processing.

Task:

- Insert 5–10 records in one go
- Use `addBatch()` and `executeBatch()`

8) Navigate through ResultSet in different directions.

Task: Scrollable ResultSet

- Move to first, last, previous, specific row
- Display records

9) Demonstrate how SQL Injection occurs and how to prevent it.

Task:

- Write login query using Statement (unsafe)
- Then rewrite using PreparedStatement (safe)

10) Fetch a student record based on user-provided ID.